Digital Health Technologies for Dementia Care: A Systematic Literature Review

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As global populations age, dementia has emerged as an escalating public health crisis, increasing the already present burdens on healthcare systems and creating new issues with informal caregivers worldwide. Traditional diagnostic pathways, which rely heavily on subjective clinical recall and periodic in-clinic evaluations, frequently fail to capture the earliest, subtle functional shifts associated with cognitive decline, resulting in delayed interventions. To address this limitation, this systematic literature review examines the integration of continuous digital monitoring, ambient intelligence, and objective digital phenotyping for early detection and longitudinal health assessment. To address this limitation while maintaining an engineering-focused approach aligned with our technical background, this systematic literature review examines the integration of continuous digital monitoring, ambient intelligence, and objective digital phenotyping for early detection and longitudinal health assessment, prioritizing technological frameworks over clinical domain considerations.

By evaluating some of the current landscape of unobtrusive in-home tracking and remote measurement platforms, this study explores how non-invasive data streams can quantify daily behavioral patterns and functional trajectories in real-world environments. The synthesis of recent evidence demonstrates that continuous digital biomarkers offer a promising pathway to bridge the gap between home-based living and clinical decision-making, providing metrics that correlate with disease progression and risk stratification. However, our critical analysis reveals persistent challenges, including dataset reliability, domain-shift vulnerabilities in unconstrained environments, patient adherence barriers, and strict privacy regulations. Ultimately, while passive monitoring frameworks hold transformative potential for proactive dementia care, future translational efforts must prioritize robust real-world validation, privacy-preserving data architectures, and seamless clinical integration.

CCS Concepts: • **Applied computing** → **Health informatics**; • **Computing methodologies** → *Machine learning*; • **Computer systems organization** → *Embedded systems*; • **Networks** → Sensor networks.

Additional Key Words and Phrases: dementia care, digital health technologies, wearable sensors, biomarkers, smart homes, systematic literature review

1 INTRODUCTION

1.1 Background and Clinical Challenges of Dementia

Dementia, including Alzheimer's disease and related neurocognitive disorders, represents one of the most critical public health challenges of the 21st century. As global populations age, the socioeconomic and healthcare burdens associated with cognitive decline continue to rise exponentially. Traditional diagnostic and monitoring pathways, which typically rely on subjective clinical recall, periodic in-clinic neuropsychological testing, and observer-dependent questionnaires, often fail to capture the earliest, subtle preclinical shifts in cognitive and functional status. As a consequence, clinical interventions frequently occur only after significant neurodegeneration has already taken place limiting potential treatment and patient wellbeing [19].

1.2 Evidence from Current Frameworks and Multicenter Initiatives

Recent large-scale initiatives and clinical studies have increasingly explored digital characterisation as a non-obtrusive approach to the early detection of cognitive decline. Unlike traditional clinical assessments, digital phenotyping utilises wearable devices and ambient sensing technologies to continuously capture objective digital biomarkers of everyday behaviour in real-world settings. While wearable sensors provide valuable insights into kinematic and physiological patterns, including gait, sleep, and cardiac activity, passive smart-home technologies offer complementary information on Activities of Daily Living (ADL) and Instrumental Activities of Daily Living

(IADL). By continuously monitoring behavioural patterns, both approaches identify subtle deviations from individual baselines that precede cognitive impairment. Aggregating these longitudinal data across sufficiently large pools supports the development of reliable statistical models for early detection

On the ambient side, foundational testbed architectures such as the Centre for Advanced Studies in Adaptive Systems (CASAS) smart-home infrastructure [3] have established robust methodologies for recording continuous behavioural patterns and detecting micro-changes in daily routines. Furthermore, multicentre European initiatives such as the RADAR-AD [17] project demonstrate the feasibility of employing remote measurement technologies to monitor functional decline in complex daily tasks among patients with Mild Cognitive Impairment (MCI) and early Alzheimer's disease. Similarly, pioneering platforms such as ORCATECH [2] investigate how non-obtrusive, in-home sensor networks can model everyday behavioural routines and detect subtle transitions from healthy cognition to MCI. Advanced computational frameworks, including deep-learning-driven models such as HEALNET [3], further exploit these rich home-telemetry streams to predict cognitive risk while reducing the burden of repeated clinical assessments.

On the wearable side, inertial sensors have been used to distinguish older adults with cognitive disorders from cognitively normal peers based on gait speed and step-time variability, achieving diagnostic performance comparable to the Mini-Mental State Examination (MMSE), a widely used clinical assessment tool [10]. Wearable sleep monitoring, combining single-channel EEG and accelerometry with artificial intelligence [13], has also been shown to distinguish patients with Alzheimer's disease from cognitively intact older adults in both hospital and home settings. Consumer smartwatches have further been proposed as passive platforms for NeuroSense, integrating gait, sleep, and cardiac data [8] within commercially available smartwatches.

Beyond proof-of-concept frameworks, several studies have evaluated passive sensing in real-world cohorts with longitudinal follow-up and clinically relevant outcomes. In the UK Biobank, gait, running, and sleep biomarkers derived from one week of wrist-worn accelerometry in more than 47,000 dementia-free adults predicted incident dementia over a median follow-up of 7.5 years, with performance comparable to established risk scores [6]. In the ORCATECH cohort, passive infrared sensors revealed that the transition to MCI was associated with deviations in room-level activity from an individual's baseline [2]. In the Rush Memory and Aging Project, a deep-learning model combining wrist accelerometry with a limited set of clinical features was proposed to estimate the five-year risk of Alzheimer's disease [16]. Together, these studies illustrate the complementary potential of population-scale wearable monitoring, longitudinal ambient sensing, and predictive modelling. They are examined in greater detail in the Clinical Studies section.

1.3 Objectives and Scope of the Review

Despite these technological strides, a critical synthesis is required to evaluate how passive smart-home data, ambient sensing and wearable-based digital biomarkers can specifically support early detection and longitudinal monitoring. This review aims to systematically examine the current evidence surrounding smart-home technologies and ambient ADL observation, as well as passively collected wearable data, providing a structured analysis of how continuous, non-invasive digital biomarkers can bridge the gap between home-based environments and clinical decision-making. Particular attention is given to what the evidence actually shows: studies on real cohorts with clinical outcomes are examined in terms of design, data collected, algorithms, performance and limitations, and are distinguished from proof-of-concept frameworks evaluated on synthetic or benchmark data.

2 REVIEW METHOD

To gather the relevant literature for this targeted review, a structured search methodology was designed to identify peer-reviewed studies examining the intersection of smart-home technologies, ambient sensing, and cognitive decline, with a specific emphasis on passive monitoring of ADL.

2.1 Search Terms and Keywords

The search strategy combined terms related to the clinical context (dementia, MCI), the technological infrastructure (smart homes, ambient sensors, wearable), and the analytical focus (digital biomarkers, passive ADL monitoring) to identify relevant evidence.

2.2 Repositories and Sources Searched

To ensure comprehensive coverage of both the medical/clinical validation and the engineering/technological frameworks, literature was searched across major multidisciplinary and domain-specific databases and repositories:

- **Google Scholar:** As a comprehensive academic search engine to capture cross-disciplinary literature, pre-prints, and hard-to-find technical reports or pilot studies.
- **PubMed / MEDLINE and ScienceDirect :** For peer-reviewed clinical evidence, medical trials, and health-focused digital biomarker research.
- **IEEE Xplore and ACM Digital Library:** For technical architectures, Internet of Things (IoT) frameworks, and ambient sensor deployments.
- **Open-Access Repositories (e.g., Zenodo, PMC):** To source benchmark datasets and methodological reports associated with smart-home testbeds.

2.3 Search and Selection Method

The search process followed a targeted screening workflow. Records retrieved from the databases were initially filtered by title and abstract to isolate studies meeting specific criteria: (1) focus on human subjects or clinical cohorts with MCI, early Alzheimer's disease or dementia; (2) deployment of unobtrusive digital sensing, either ambient or smart-home sensors (such as PIR detectors and contact sensors) or worn wearables (such as wairt-worn IMU, smartwatches, etc ...); (3) concentration on objective digital biomarkers, automated classification or screening outcomes, such as ADL, gait, or sleep measures; and (4) studies showcasing technical knowledge and engineering challenges to maintain a compatible background.

Articles were excluded if they focused purely on human-computer interaction or social adoption, or if they relied exclusively on in-clinic neuropsychological testing, neuroimaging, or fluid biomarkers without a digital sensing component. Furthermore, given the project itself, we decided to focus our research on 3 main topics depicted as the 3 main technologies used in this area. More details are explained in section 4.

3 SIGNIFICANCE AND IMPACT OF THE AREA

Dementia represents one of the most critical global health and socioeconomic challenges of the 21st century. According to the World Health Organization (WHO) [23], just in 2021, over 55 million people worldwide were living with dementia, a number projected to nearly triple to 152 million by 2050 due to accelerating global population aging [11]. Approximately 10 million new dementia diagnoses occur annually [23]. However, this figure likely misrepresents the issue we are currently facing, as a total of 75% of dementia cases worldwide remain undiagnosed [5].

The economic burden associated with the disease is immense. In 2019, the global societal cost of dementia was estimated at $1.3 trillion, a figure expected to exceed $2.8 trillion by 2030 if adjusted for inflation [1]. In the United States alone, total direct healthcare and long-term care costs reached $345 billion in 2023 [4]. Beyond formal healthcare, the largest social impact derives from informal caregiving: unpaid care provided by family members and friends. Globally, informal caregivers contribute an estimated 82 billion hours of care annually, equivalent to approximately 40 million full-time jobs, leading to profound productivity loss and significant physical, emotional,

and financial strain [22]. Furthermore, patients suffer a severe loss in Disability-Adjusted Life Years (DALYs), with dementia accounting for over 28 million DALYs globally [1].

Given the difficulties in treating dementia-related conditions, the effectiveness of therapeutic and non-pharmacological interventions, as well as care planning, relies heavily on the timing of intervention. Early and accurate diagnosis enables timely clinical management, reduces avoidable hospitalizations, and potentially slows cognitive decline [19]. Cognitive impairment is often underdiagnosed or identified late, partly due to limitations in routine clinical assessment. This review examines current approaches to early detection and management, focusing on digital health technologies, AI-based screening, assistive technologies, and specialized care models.

4 TECHNOLOGY

Digital health technologies for dementia care and early detection contain a broad spectrum of solutions, ranging from non-invasive consumer-grade hardware to sophisticated data analytics frameworks. To provide a comprehensive overview of both commercial off-the-shelf (COTS) products and research prototypes, the literature is categorized into a three-tiered taxonomy based on the sensing environment and analytic focus:

(1) **Wearable Devices & Kinematic Sensing:** Wearable solutions focusing on individual physiological and movement tracking.
(2) **Smart-Home & Ambient Sensing:** Unobtrusive in-home environments designed to monitor daily behavioral patterns.
(3) **Longitudinal Monitoring & Predictive Algorithms (Digital Biomarkers):** Software and analytical frameworks leveraging continuous long-term data stream analysis to model disease trajectory.

4.1 Wearable Devices & Physiological/Kinematic Sensing

Wearable technology, including body-worn inertial sensors, wearable EEG devices, and smartwatch-based devices, enables continuous assessment of motor and physiological function in real-world settings. These devices capture high-frequency kinematic data that ambient sensing cannot provide, including gait parameters (cadence, speed, step time variability, asymmetry), balance metrics, and sleep architecture. Waist-mounted accelerometers and gyroscopes are economical for gait analysis during standardized mobility tasks. While compact wearable EEG sensors offer superior comfort and home usability compared to laboratory-based polysomnography, they do sacrifice signal fidelity due to fewer electrodes and motion artifacts. Consumer smartwatches and wrist-worn accelerometers (e.g., Fitbit, Apple Watch) provide accessible longitudinal tracking of gait, heart-rate variability, and sleep metrics. These kinematic and physiological markers serve as early digital biomarkers for detecting subtle motor and cognitive decline before obvius clinical manifestations emerge.

4.2 Smart-Home & Ambient Sensing

Ambient intelligence technologies allow for continuous and non-intrusive monitoring of individuals within their natural living environments. This domain relies on Passive Infrared (PIR) motion sensors, smart mats, magnetic door contact sensors, and appliance energy consumption monitoring. By aggregating these environmental data points, systems can construct baselines for ADL. These systems automatically detect subtle anomalies, such as nocturnal wandering, changes in room transitions, or functional decline, indicative of MCI and early dementia.

4.3 Longitudinal Monitoring & Predictive Algorithms

Complementing hardware-centric approaches, this area focuses on the computational layer that processes prolonged, multi-source digital streams. By applying advanced algorithmic models and machine learning pipelines

to long-term data, these studies aim to extract meaningful digital biomarkers. Rather than relying on cross-sectional assessments, longitudinal predictive algorithms model disease progression over months or years, facilitating early prediction, risk stratification, and timely clinical intervention.

5 CLINICAL STUDIES

5.1 Gait Feature–Based Model for Classifying Cognitive Disorders Using a Single Wearable Inertial Sensor

Jeongbin Park, Hyang Jun Lee, Ki Woong Kim and colleagues developed a gait-based model to classify older adults with CD using one wearable inertial [18].

Aims. The study asked whether gait speed and temporal gait variability, captured by a single sensor, can accurately separate older adults with CD (MCI or dementia) from cognitively normal controls, and whether this matches an MMSE-based model.

Study Design. A quantitative, cross-sectional study. Participants (n=595) were randomly split into development (80%) and validation (20%) sets, and a logistic regression with forward selection was compared against an MMSE-based model.

Study Methods. Participants came from the population-based from the Korean Longitudinal Study on Cognitive Aging and Dementia cohort: 6,818 residents of Yong-In were invited and 595 were eligible after excluding stroke, Parkinson's disease and other movement disorders, knee/hip/ankle surgery, and spinal disease. Diagnoses were confirmed by a geriatric psychiatrist panel after MMSE, and a structured interview. A waist-worn IMU (L3–L4, 250 Hz) recorded three walks along a 14 m walkway at a comfortable pace, and the central 10 m was analyzed.

Study Results. Of 595 participants, 101 (17%) had CD (94 MCI, 6 AD dementia, 1 dementia with Lewy bodies). The CD group was older (75.7 vs. 72.7 years), less educated, slower (101 vs. 115 cm/s), and had higher step-time variability (4.0% vs. 3.2%). The model retained gait speed, variability and education, reaching an AUC of 0.788 in development and 0.811 in validation, versus 0.762 and 0.741 for the MMSE model (differences not significant), with higher sensitivity but lower specificity than MMSE.

Conclusions. The authors conclude that single-sensor gait analysis is a promising, easily repeatable marker of CD. Limitations include the lack of external validation, the all-cause CD target, the cross-sectional design, and the inclusion of education in the "gait" model.

5.2 Wearable Sensor-Based MCI Identification: A Multi-Domain Gait Analysis Approach With Association Rule Mining

Keyi Huang, Ruiyuan Wu, Xinyu Liu, Liao Jing and Yang Zhao proposed an interpretable framework for identifying MCI from a waist-worn IMU during a standard physical-performance test [14].

Aims. The study aimed to identify MCI without extra screening tasks, compare time, frequency, time-frequency and nonlinear features and their fusion, and derive transparent rules for clinical use.

Study Design. A quantitative, cross-sectional case-control study evaluated with 50 repeated stratified 80/20 holdout splits .

Study Methods. Adults aged 60 or older were recruited from community hospitals in Guangzhou (11/2023–04/2025). Those with other neuropsychiatric disorders or severe mobility limitations were excluded. MCI required subjective decline (AD8 $\geq$ 2), education-adjusted abnormal MMSE, preserved basic ADL, and no dementia (DSM-5). An IMU at L3–L5 recorded the seven Short Physical Performance Battery tasks (balance, 3 m walk, chair stands

illustraded at 1) at 10 Hz. Features were selected by statistical filtering, regression and SVM-based backward selection, and fed to five classifiers; K-means discretization and Apriori produced association rules.

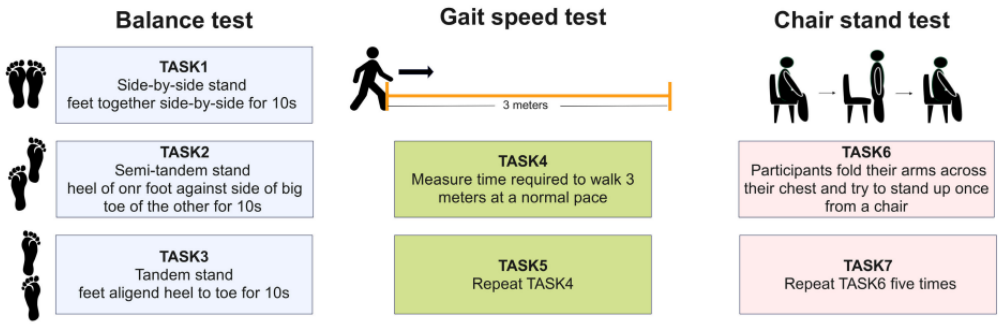


Fig. 1. Short schema of the application of IMUs to the Short Physical Performance Battery (SPPB) Assessment for MCI Detection. Adapted from [14].

Study Results. The 201 participants (71 MCI, 130 controls) did not differ in age, sex, education or SPPB score, while MMSE differed (22.7 vs. 27.2). The fused feature set with an SVM reached an AUC of 0.897 and an accuracy of 0.828; the best single domain (time-frequency) reached an AUC of about 0.79. Nonlinear features dominated the 23 selected features. The top ten rules had confidence 0.81–0.88 and lift 2.29–2.48, centered on reduced entropy of trunk angular velocity in balance tasks. Dropouts are not reported.

Conclusions. The authors conclude that the framework offers a scalable, interpretable pre-diagnostic screening tool.

5.3 Wearable Sleep Recording Augmented by Artificial Intelligence for Alzheimer's Disease Screening

Elisabeth Heremans, Astrid Devulder, Maarten De Vos and colleagues tested whether wearable sleep recordings, processed fully automatically, can screen for Alzheimer's disease [13].

Aims. The study had three aims: automate sleep staging from a one-channel EEG and accelerometry wearable; discriminate AD patients (including prodromal AD) from cognitively intact elderly (CIE); and identify the most informative sleep features.

Study Design. A quantitative, cross-sectional proof-of-concept study pooling two datasets: in-hospital recordings of older adults with suspected sleep apnea and home-based recordings of AD patients and controls. Manually scored PSG was the ground truth, with AI-scored PSG as a comparator.

Study Methods. Of 147 participants, 45 trained the sleep-staging model and 102 formed the test set (35 AD, 67 CIE). AD was clinically probable (7) or biomarker-proven (28); 12 with MMSE $\geq$ 27 were labeled prodromal. Controls were healthy volunteers or hospital patients without recorded cognitive complaints (those with complaints were excluded). The studies had approval from the University Hospitals Leuven ethics committee, consent from participants or caregivers, and registered trials (NCT04755504, NCT03617497). Participants wore a behind-the-ear wearable (SensorDot) during overnight PSG.

Study Results. AD patients were older (73–76 vs. about 68 years). Wearable staging reached 65.5% accuracy (κ = 0.498) versus 76.9% for PSG. AD detection on the wearable reached an AUC of 0.90 (0.76 for prodromal AD), similar to AI-scored PSG (0.88/0.77) and slightly below manually scored PSG (0.94/0.79). Hypnogram features

performed far worse (AUC 0.61–0.64) and near chance for prodromal AD; OSA did not change performance. Key features reflected EEG slowing in wake, light sleep and REM.

Conclusions. The authors conclude that one-channel EEG with accelerometry holds enough information for AD screening and that perfect sleep staging is unnecessary. Limitations: heterogeneous control definitions without biomarkers (preclinical AD cannot be excluded), older AD patients, all AD recordings made at home while many controls were recorded in hospital, and only 12 prodromal cases.

5.4 NeuroSense: Smartwatch-based Early Detection Framework for Alzheimer's Disease

Aarju Dixit and Debasis Das proposed NeuroSense, a framework for passive early detection and staging of Alzheimer's disease from consumer smartwatch [8].

Aims. The study aimed to provide a non-invasive and scalable alternative to neuroimaging and cerebrospinal fluid analysis by passively monitoring gross motor function (walking speed, stairs), fine motor control (eating), circadian rhythm (sleep duration, count, and posture), and social behavior (mobile application usage) with the IMU and PPG sensors of Apple Watch and Fitbit devices. The authors also aimed to overcome the low adherence to periodic testing protocols.

Study Design. The work is a computational, simulation-based study with no human participants. The dataset was "synthetically generated using artificial intelligence techniques" to mimic AD-related features, and was split 80%/20% into training and testing subsets. The task is a five-class classification of disease stage: not demented, MCI, early-stage demented, partially demented, and demented.

Study Methods. Because the data are synthetic, no inclusion or exclusion criteria or recruitment. The pipeline imputes missing values with the feature mean, applies time filtering and segmentation, and trains two models: a one-hidden-layer shallow neural network in which features are rescaled by a "cross-sectional entropy" term, and a modified 1D SqueezeNet with deeper fire modules and larger filters, trained for 50 epochs. A majority-voting mechanism aggregates predictions across monitoring sessions, and a cloud back-end is proposed for continuous model updating.

Study Results. The reported dataset comprises 6264 simulated subjects (3117 not demented, 1281 demented, 647 MCI, 624 partially demented, 595 early-stage demented) monitored for 21 days. The authors report a detection accuracy of 94.4%, higher than that of earlier approaches listed in their comparison table (e.g., 42.3% for MCI with actigraphy and 88.9% for MCI with audio, depth, and radar sensing). In the synthetic data, the not-demented group sleeps mostly 7–8 hours, while demented and partially demented groups concentrate around 6–7 hours; the median walking speed is about 1.2 m/s for not demented, below 1.0 m/s for MCI, and 1.3–1.4 m/s for the early-stage, partially demented, and demented groups.

Conclusions. The authors conclude that NeuroSense offers a practical tool for passive, real-world AD monitoring and plan to add electrodermal activity and ambient light sensing, as well as federated learning for privacy. For the purposes of this review, however, the evidence is limited. All results come from synthetic data, so they demonstrate the computational pipeline rather than clinical validity; the accuracy is not comparable with the cited prior studies, which used real cohorts, different targets, and different sensors; the reported training and validation accuracy curves (in the range 0.10–0.35) do not match the stated 94.4%; the synthetic walking speed is higher in the demented stages than in non-demented subjects, contrary to the slower gait observed in real cognitive-disorder cohorts; and no per-class metrics, external validation, or real-world data are provided. NeuroSense is therefore best read, like HEALNET, as a proof-of-concept framework that motivates future validation on real participants.

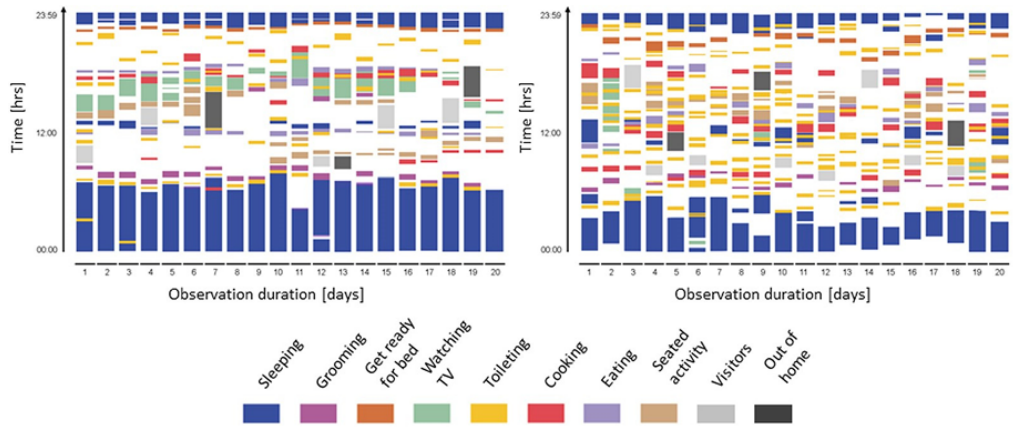


Fig. 2. Activity maps of a healthy control (left) and a patient with dementia (right) visualized from data continuously measured for 20 consecutive days. Activity maps of patients with dementia reveal unorganized behavior patterns, and heterogeneity differed significantly between the healthy control and the patient.[24].

5.5 Nonwearable Sensor-Based In-Home Assessment of Subtle Daily Behavioral Changes as a Candidate Biomarker for MCI

In this review, Takao Yamasaki and Shuzo Kumagai evaluated the assessment of daily behavioral changes in patients with MCI and Alzheimer's disease (AD) using nonwearable sensor-based in-home systems [24].

Aims. The study aimed to investigate whether tracking daily behavioral patterns via digital technologies could replace or assist traditional assessment methods for the early detection of MCI and AD. The core hypothesis posited that subtle behavioral shifts captured by nonwearable in-home sensors can serve as early digital biomarkers.

Study Design. The paper was structured as a comprehensive review and qualitative-analytical assessment of existing literature, comparing traditional performance- and questionnaire-based methods with digital nonwearable approaches across healthy controls, MCI subtypes, and dementia patients.

Study Methods. The authors analyzed previous research utilizing various nonwearable sensor technologies installed in naturalistic home environments, such as passive infrared (PIR) motion sensors, magnetic contact door switches, vibration sensors, pressure floor mats, and appliance power consumption monitors. The synthesized studies tracked activities over durations ranging from 20 consecutive days to over 3 years, examining walking speed, sleep interruptions, time spent out-of-home, and daily routines.

Study Results. The review demonstrated that nonwearable sensors successfully captured quantifiable differences between healthy individuals and those with cognitive impairment. Specifically, patients with MCI exhibited greater day-to-day activity variability, altered walking speeds, and increased indoor time. As shown through continuous long-term monitoring, these distinctions are clearly visible in daily routines: while healthy controls display structured patterns, activity maps of dementia patients reveal unorganized and fragmented behaviors, as illustrated in Figure 2. Furthermore, machine learning models (like support vector machines and random forests) applied to these sensor data streams achieved high predictive performance, detecting MCI with ROC curve scores up to 0.97.

Conclusions. The authors concluded that nonwearable sensor-based in-home assessments offer an objective, quantitative, and realistic method to capture early behavioral shifts, representing viable early diagnostic biomarkers for MCI and AD. Despite challenges regarding cost, installation effort, and data standardization, such continuous monitoring systems hold significant potential for point-of-care testing and personalized medicine in dementia care.

5.6 Remote Monitoring Technologies in Alzheimer's Disease: Design of the RADAR-AD Study

Marijn Muurling, Casper de Boer, and colleagues designed the RADAR-AD (Remote Assessment of Disease and Relapse – Alzheimer's Disease) study to evaluate the accuracy and validity of remote monitoring technologies (RMTs) in measuring functional decline across the preclinical-to-moderate stages of Alzheimer's disease (AD) [17].

Aims. The study aimed to assess whether RMTs—such as smartphone applications, wearables, and home-based sensors—can accurately measure functional decline in a real-world environment with increased sensitivity compared to standard clinical rating scales. The study aimed to assess whether RMTs, such as smartphone applications, wearables, and home-based sensors, can accurately measure functional decline in a real-world environment. The researchers compared their sensitivity to standard clinical rating scales. Specifically, the project addressed three core aims: differentiating between disease stages, conducting head-to-head comparisons with clinical standards, and evaluating the technical feasibility and user acceptability of RMTs.

Study Design. The study was structured as a multicentre, observational, cross-sectional cohort study comprising three tiers: a main study using smartphone apps and wearables ($n = 220$), a first sub-study adding fixed home sensors ($n = 40$), and a second sub-study observing participants in a controlled smart home environment ($n = 10$). Participants were evenly distributed across 13 European countries and classified into four groups based on biomarkers, MMSE, and Clinical Dementia Rating (CDR) scores: healthy controls, preclinical AD, prodromal AD/MCI, and mild-to-moderate AD.

Study Methods. The study operationalized continuous and periodic monitoring over 8 weeks using a robust suite of RMTs deployed via the RADAR-base platform. Continuous tracking was performed using wrist-worn accelerometers (Fitbit Charge 3 and Axivity AX3) and a passive smartphone app (pRMT) collecting mobility and social communication data. Periodic assessments included active smartphone tasks (Mezurio app), augmented reality cognitive tests (Altoida MD), digital financial management apps, and an optional wearable camera (Autographer) to contextualize daily activities. As illustrated in Figure 3, the multi-sensor setup strategically combines continuous background logging with periodic active evaluations to capture fine-grained behavioral changes in the participant's natural environment. Sub-studies incorporated residential movement sensors, sleep headbands (DREEM), and vehicle driving loggers (CANedge). Ethical approvals were secured across all sites, and data privacy was strictly managed under GDPR [9] protocols.

Study Results. As a protocol and design paper, the primary focus centered on establishing the methodological framework and recruitment targets across 14 clinical sites in 13 countries to enroll 220 well-characterized participants with confirmed amyloid-β pathology. The analytical plan established mixed-effect models for repeated measures (MMRM) and multivariate machine learning classifiers (such as random forests) to process high-frequency sensor streams, map digital biomarkers to functional domains (such as the Amsterdam IADL questionnaire), and differentiate between clinical AD stages.

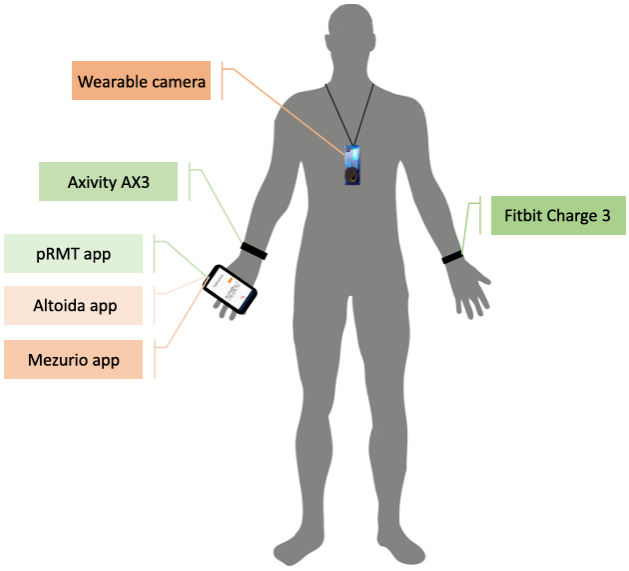


Fig. 3. Representation of the smartphone applications and wearable devices used in the RADAR-AD main study. Devices in green measure continuously, while devices in orange measure periodically. [17].

Conclusions. The authors concluded that the RADAR-AD study is well positioned to establish the clinical utility of RMTs for objective, continuous, and ecologically valid functional monitoring in real-world environments. Successful validation of these digital endpoints holds strong potential to improve clinical trial sensitivity, accelerate drug development, and enhance personalized care for individuals affected by Alzheimer's disease.

5.7 Smart Home Internet of Things-Based Behavioural Analysis for Early Detection of Cognitive Decline: Toward Saudi Future Vision

Arij Alfaidi, Shadi Majed Alshraah, and colleagues proposed HEALNET (Home Environment Assisted Learning Network), a hybrid framework designed to analyze continuous daily behavioral patterns from smart-home IoT data for early cognitive-risk screening [3].

Aims. The study aimed to develop a non-invasive, data-driven framework that analyzes longitudinal behavioral patterns derived from smart-home Internet of Things (IoT) sensors to support early cognitive-risk screening rather than direct clinical diagnosis. The core objective was to bridge the gap between episodic, subjective clinical assessments and continuous real-world monitoring by effectively integrating spatial, temporal, and statistical behavioral representations.

Study Design. The study was structured as an experimental and data-driven computational research framework evaluated using publicly available benchmark datasets. Rather than recruiting human patient cohorts directly for a clinical trial, the authors utilized the CASAS smart-home datasets from the University of Washington, partitioning the data into training (70%), validation (15%), and testing (15%) subsets to evaluate hybrid machine learning and deep learning models.

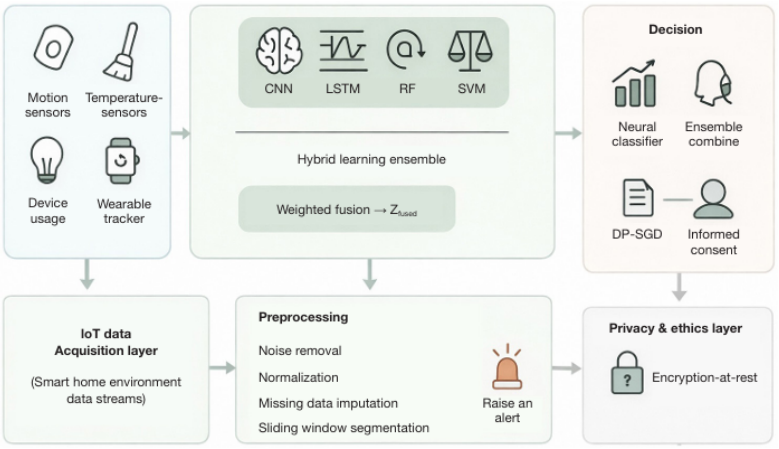


Fig. 4. Proposed architecture of the HEALNET framework for cognitive decline detection using IoT-based behavioural data.[3].

Study Methods. The methodological pipeline operationalized multimodal IoT sensor streams—including passive infrared (PIR) motion detectors, temperature sensors, door switches, and appliance usage logs—capturing time-stamped activity events over extended durations. Raw event logs underwent pre-processing involving noise removal, timestamp alignment, normalization, and sliding-window segmentation. Feature engineering extracted activity durations, room transition frequencies, sleep consistency, and appliance interaction patterns. These features fed into HEALNET's hybrid architecture: a convolutional neural network (CNN) encoder for spatial feature extraction, a long short-term memory (LSTM) network for temporal sequence modeling, and ensemble classifiers including Random Forest (RF) and support vector machines (SVM). As illustrated in Figure 4, the framework integrates these multimodal streams through an attention-based fusion layer while incorporating differential privacy mechanisms (DP-SGD) to safeguard participant data.

Study Results. The experimental evaluation demonstrated that the proposed HEALNET framework achieved a classification accuracy of 94.2%, outperforming baseline single-model approaches such as standalone SVM (85.7%), Random Forest (88.5%), and LSTM (91.2%). Confusion matrix analysis confirmed high true-positive recognition rates for behavioral patterns consistent with MCI, while correlation analysis verified strong associations between elevated cognitive-risk patterns and indicators such as sleep irregularity and reduced activity levels.

Conclusions. The authors concluded that continuous, unobtrusive behavioral monitoring using smart-home IoT data provides reliable indicators for cognitive-risk screening. Serving as a research-stage framework for data-driven behavioral analysis rather than a clinical diagnostic tool, HEALNET aligns with digital health innovation objectives by offering a scalable, privacy-aware, and ethically grounded approach to proactive cognitive health monitoring.

5.8 Unobtrusive Detection of Mild Cognitive Impairment (MCI) in Older Adults Through Home Monitoring

Ahmad Akl, Jasper Snoek and Alex Mihailidis developed a statistical model of in-home activity to detect the transition to MCI, using sensor and clinical data collected by ORCATECH (Oregon Center for Aging & Technology) [2].

Aims. The study aimed to detect the transition from healthy cognition to MCI from everyday activity in the home, without any action from the resident, and to outperform an earlier method based on predefined measures. The implicit hypothesis is that the divergence of a person's room-level activity pattern from their own cognitively intact baseline signals the transition to MCI; the null hypothesis is that activity patterns do not differ between intact and MCI periods.

Study Design. A quantitative, longitudinal observational study based on continuous in-home monitoring (mean duration three years) with annual clinical assessments. The baseline was each subject's own activity distribution during cognitively intact periods, against which later time windows were compared.

Study Methods. Sixty-eight participants, recruited in the Portland (Oregon) area, were aged 70 or older, lived independently in a home larger than a studio apartment, were cognitively healthy at recruitment (CDR and MMSE within the set limits) and in normal health for their age. Ethics approval and consent are not described in the sources consulted. Passive infrared (PIR) sensors in the most used rooms sent each activation to a transmitter, which time-stamped it and stored it in an SQL database. A weekly online questionnaire (visitors, days away, health or medication changes, emergency visits) was used to discard non-representative days: annual visits, guests, absences, maintenance, limiting health problems and sensor failures. Annual home assessments (MMSE and CDR) provided the ground truth: CDR 0 meant intact cognition and CDR 0.5 meant MCI, and data were labelled as intact, in transition or MCI. Activity in four rooms (bedroom, bathroom, kitchen, living room) was modelled as independent non-homogeneous Poisson processes over 48 half-hour intervals per day. Distributions were compared through a normalized Kullback–Leibler divergence against a reference estimated from intact periods, using sliding windows, and a simple threshold on the divergence trajectory classified periods as intact or MCI. The sensors required no action from participants, apart from the weekly questionnaire, and no intervention was applied.

Study Results. Of the 68 participants, 7 men and 61 women, 15 transitioned to MCI, 2 men and 13 women. With 12-week windows, the mean AUC-ROC was 0.716 and the mean AUC-PR 0.706; by room, AUC-ROC ranged from 0.687 (bathroom) to 0.745 (living room). With half the window length, the approach exceeded the best result of the earlier method (AUC-ROC 0.57 and AUC-PR 0.43 at 24 weeks). At the same 12-week window, the earlier method scored 0.40 and 0.26, and the authors report gains of 31% and 44%, which correspond to the absolute differences. For most subjects the divergence started to rise several months before the assigned transition period, and those who transitioned spent more time in the bedroom between 23:00 and 06:00, consistent with disturbed sleep.

Conclusions. The results support the use of a simple, interpretable statistical model on passive home sensors, with moderate performance and early behavioural signals preceding the formal transition by months. The authors report four limitations: a small sample, especially of subjects who transitioned to MCI; dependence on annual assessments, which can attach wrong labels to the months before diagnosis; three subjects with no divergence increase, possibly because of amnestic MCI, which the CDR does not distinguish from the non-amnestic type; and the inability of PIR sensors to capture passive activities such as deep sleep.

5.9 Daily-Life Walking Speed, Running Duration and Bedtime from Wrist-Worn Sensors Predict Incident Dementia: A Watch Walk – UK Biobank Study

L. L. Y. Chan, M. T. Espinoza Cerda, M. A. Brodie, S. R. Lord and M. E. Taylor investigated whether passively collected wrist accelerometry can predict incident dementia in the UK Biobank cohort, using the Watch Walk method developed at Neuroscience Research Australia [6].

Aims. The study asked whether digital biomarkers of real-life walking and sleep, extracted from a wrist-worn accelerometer, are associated with incident dementia in older adults, and whether they predict it as well as established risk scores (ANU-ADRI and the UKB-Dementia Risk Score). The implicit hypothesis is that these passively acquired measures carry independent prognostic information; the corresponding null hypothesis is that they do not.

Study Design. A quantitative, longitudinal prospective cohort study. A baseline accelerometry recording of up to seven days was linked to dementia diagnoses recorded over a median follow-up of 7.5 years (IQR 7.0–9.0). Associations were tested with univariate and multivariate Cox proportional hazards models, and the multivariate model was compared with the two established risk scores, which served as the baseline.

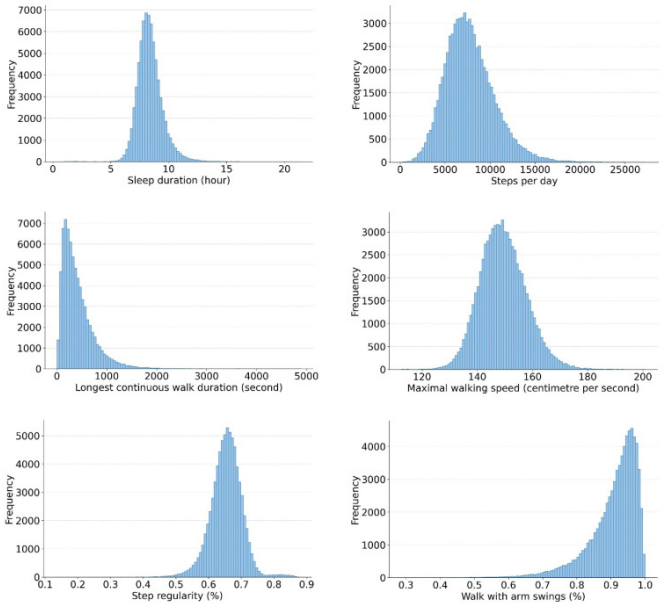


Fig. 5. Confusion matrix of stage 1 classification. [6].

Study Methods. Participants were UK Biobank members aged 60 or older and free of dementia at baseline (n = 47,371). Further exclusion criteria, recruitment procedures, ethics approval and consent are not described in the abstract and registry entries consulted. Participants wore a wrist accelerometer for up to seven days (reportedly an Axivity AX3, worn between 2013 and 2015). The Watch Walk method extracted 19 digital biomarkers from the raw signal, including real-life walking measures (peak and usual walking speed, walking duration, bout length, percentage of steps with arm swing) and sleep measures (sleep duration and bedtime). The study was purely observational: there was no intervention and no action was required from participants during data collection.

Study Results. Participants had a mean age of 67 ± 4 years and 52% were women. The distributions of the extracted biomarkers are shown in Figure 5 and the performance of the stage activity classification model is illustrated in the confusion matrix in Figure 6. During follow-up, 387 participants (0.8%) were diagnosed with dementia. Among the gait parameters, lower peak walking speed had the strongest association with dementia (32% lower risk for each standard-deviation increase in speed), followed by lower daily step count (30%) and higher step-time variability (17% higher risk). After adjustment for age and sex, running duration, peak walking

speed and an early bedtime remained independent significant predictors. The multivariate model performed comparably to ANU-ADRI and the UKB-Dementia Risk Score.

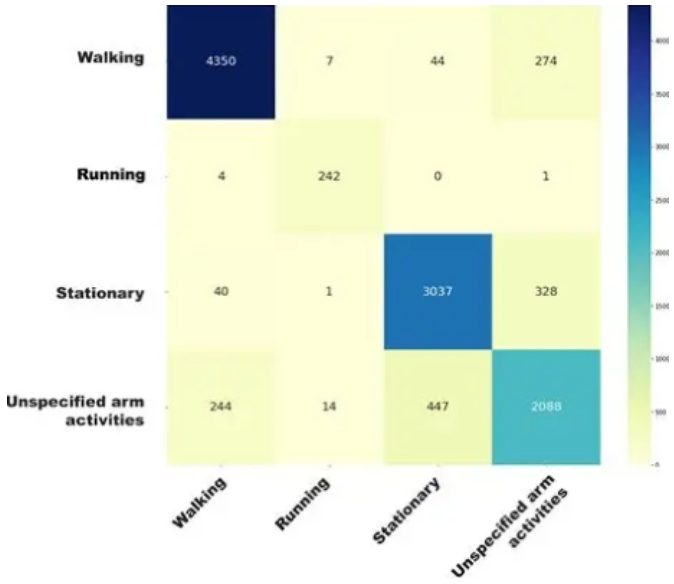


Fig. 6. Confusion matrix of stage 1 activity classification (walking, running, stationary, and unspecified arm activities). [6].

Conclusions. The study supports the use of passive wrist-sensor measures as dementia risk markers with prognostic value comparable to traditional scores. Its value lies in matching the existing tools with parameters acquired remotely and passively, not in outperforming them. Limitations are that the 7.5 years indicate the observation window and not a guaranteed diagnostic lead, that a single week of baseline recording cannot capture individual trajectories, and that only 387 cases (0.8%) make the classes strongly imbalanced.

5.10 Prediction of Incident Alzheimer's Disease in Older Community-Dwelling Adults Using Wrist Accelerometry and Deep Learning

J. Midroni, L. Yu, J. S. Rabin, X. Liu, M. Goubran, A. S. Buchman, D. A. Bennett, A. Lim and colleagues used data from the Rush Memory and Aging Project (MAP) to predict Alzheimer's disease (AD) within five years; the work is published as an abstract in the *Alzheimer's & Dementia* supplement [16].

Aims. The study aimed to predict clinical AD within five years in community-dwelling older adults by combining wrist accelerometry with a few clinical features chosen for their ease of collection. The implicit hypothesis is that the combined model outperforms models based on a single data source; the null hypothesis is that it brings no advantage.

Study Design. A quantitative predictive-modelling study on data from a longitudinal cohort. Participants were divided into a training set (875) and a test set (395). The combined neural network was compared against an accelerometer-only model and a clinical-only model, and its binarized output was then used as a predictor of five-year AD-free survival in a Cox model.

Study Methods. Participants were free of dementia at enrolment, survived, and had a known clinical AD outcome within five years of enrolment. Further inclusion and exclusion criteria, recruitment, ethics and consent are not detailed in the abstract. Participants wore a wrist accelerometer for ten days. A feedforward neural network combined accelerometer features with five clinical features: age, sex, education, social isolation and purpose in life. The study was observational with no intervention, and the authors state that tuning and validation of the model are still in progress.

Study Results. The test set had a mean age of 81.5 ± 6.8 years, and 14.4% developed clinical AD within five years. The combined model reached a sensitivity of 89%, a specificity of 70% and an F1 score of 0.87, against 67%, 70% and 0.68 for the accelerometer-only model and 76%, 80% and 0.78 for the clinical-only model. Accuracy was similar with and without MCI at baseline. In the Cox model the binarized output gave a C-index of 0.729 (95% CI 0.69–0.77), a hazard ratio of 6.90 (95% CI 4.15–11.47) and a log-likelihood ratio of 71.05.

Conclusions. The multimodal model improves sensitivity over either single source, which supports the idea of combining sensor and clinical data. The specificity of 70% equals that of the accelerometer-only model and is lower than that of the clinical-only model, so the result is a trade-off rather than an excellent performance. With a prevalence of 14.4%, an F1 of 0.87 suggests a class-weighted average, a detail the abstract does not clarify. Since this is a conference abstract and validation is ongoing, the results need confirmation in the full paper.

6 CRITICAL ANALYSIS

From an engineering perspective, deploying digital health technologies for dementia care involves complex trade-offs between data granularity, computational overhead, privacy constraints, hardware ergonomics, and architectural scalability. Evaluating the reviewed literature reveals several critical engineering challenges and technical deficits.

6.1 Architectural Trade-offs: Edge vs. Cloud and Sensor Granularity

A fundamental architectural challenge lies in balancing sensor data resolution with processing and bandwidth costs:

- **Wearable Kinematic and Physiological Sensors** (e.g., IMUs, 1-channel ECG/EEG) capture high-frequency time-series data (often 10Hz to 250Hz), requiring robust edge-filtering and noise reduction (motion artifact mitigation). While frameworks like NeuroSense propose continuous cloud telemetry, streaming raw high-frequency data from consumer devices poses severe bottlenecks regarding battery consumption, local storage, and network bandwidth in real-world deployments.
- **Smart-Home IoT Infrastructures** (e.g., PIR motion detectors, magnetic contact switches) adopt an event-driven, low-bandwidth model. While highly efficient for long-term monitoring, they provide limited interpretability of the underlying physiological mechanisms. Simple event streams cannot distinguish between complex overlapping activities or subtle motor deficits without complex probabilistic models and an ample data pool to extract meaningful correlations.

6.2 Algorithmic Limitations and Dataset Reliability

The algorithmic models applied to smart-home and ambient telemetry streams show notable technical and methodological vulnerabilities that interfere with their clinical translation:

- **The Synthetic vs. Real-World Data Divide:** Many state-of-the-art frameworks (such as HEALNET and NeuroSense) rely heavily on heavily pre-processed benchmark datasets (e.g., CASAS) or synthetic data. While these standardized datasets allow rapid prototyping of complex deep-learning architectures—such as Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) hybrid fusion networks- they

fundamentally fail to capture the chaotic, highly unpredictable nature of human behavior in progressive neurodegenerative pathologies. Consequently, synthetic or controlled models sometimes yield counter-intuitive or contradictory correlations that cannot be used in natural settings once publicly available.

- **Domain Shift and Ground-Truth Scarcity:** Machine learning pipelines trained on specific demographic cohorts, single-site smart-home testbeds, or rigidly controlled experimental environments suffer from severe domain shift when deployed in unconstrained domestic settings with diverse architectural layouts, multi-occupants, and pets. Compounding this generalizability barrier is the critical scarcity of fine-grained, longitudinal clinical ground-truth labels ~~(such as episodic CSF/PET biomarkers or formal clinician-administered CDR/MMSE scores)~~; most models rely instead on heuristic activity annotations rather than verified neurocognitive markers.
- **Sensor Noise and Hardware Drift:** Unlike controlled clinical trials, longitudinal in-home deployments are inherently vulnerable to hardware artifacts, dropped wireless packets, dead batteries, unworn devices, and uncorrected temporal drift, which can severely degrade predictive accuracy if models lack robust noise-filtering mechanisms.

6.3 Low User Adherence and Wearable Compliance Barriers

While wearable sensors offer rich physiological and kinematic data streams, their real-world deployment faces severe practical limitations stemming from user adherence challenges. This is a known issue in dementia research. Prospective memory challenges, such as remembering to charge devices, press buttons at scheduled times, or change wearable devices, substantially reduce sustained usage among people with dementia [12]. Beyond cognitive barriers, device design and perceived usefulness significantly influence acceptance [21]. The appearance and social visibility of wearables also create emotional resistance, particularly among cognitively intact or mildly impaired individuals. The implications for data validity are profound: high adherence in cognitively preserved populations produces systematically biased datasets with low representative value, compromising clinical validity and generalizability. Critically, the relationship between missing data and cognitive decline cannot be addressed through advanced imputation methods alone [12].

6.4 Data Governance, Privacy, and Scalability

Continuous passive monitoring of vulnerable populations generates profound security and privacy engineering challenges under strict regulatory frameworks like GDPR [7]. Storing continuous location, sleep patterns, and behavioral telemetry on centralized cloud servers introduces single points of failure and high vulnerability to data breaches. Future engineering efforts must shift from centralized machine learning toward Federated Learning (FL) [15] and differential privacy mechanisms ~~(such as DP-SGD implemented in advanced IoT frameworks)~~, enabling decentralized model training directly on local edge gateways while preserving patient data sovereignty and minimizing transmission overhead.

6.5 Caregiver Burden, Ecological Validity, and Surveillance Ethics

While ambient and passive sensing frameworks are frequently promoted as solutions to reduce caregiving strain, a critical evaluation reveals significant psychological and social complexities regarding their real-world deployment.

- **The Surveillance Dilemma:** Continuous in-home monitoring inherently walks a fine line between safety and privacy intrusion. Constant behavioral tracking can provoke feelings of loss of autonomy and psychological distress among older adults [20].
- **Caregiver Reliance and False Security:** Conversely, caregivers often face an indirect burden. Rather than alleviating stress, opaque algorithmic alert systems can generate anxiety or foster a misplaced sense of security, causing family members to overlook subtle clinical symptoms while relying blindly on automated

anomaly flags. Furthermore, the active participation burden (such as managing app schedules or completing weekly IADL questionnaires, as seen in RADAR-AD) heavily relies on caregiver adherence, a variable that is rarely accounted for in controlled technical evaluations.

6.6 Synthesis and Methodological Trends across Reviewed Studies

As synthesized in Table 1, the reviewed literature reflects a clear methodological evolution over recent years with a strong shift from small-scale, cross-sectional ambient prototypes to massive, multicentric prospective cohorts and advanced machine learning validation.

- **Sample Size and Validation Gap:** While early studies (e.g., Yamasaki & Kumagai, Akl et al.) relied on small cohorts ($n < 70$) to prove foundational ambient sensing feasibility, recent large-scale initiatives like Chan et al. ($n = 47,371$) and Park et al. ($n = 595$) demonstrate a strong industry push toward population-level validation. However, a persistent gap remains between simulation-based pipelines (e.g., Dixit & Das with synthetic data) and unconstrained real-world deployment.
- **Accuracy vs. Ecological Validity Trade-off:** Controlled smart-home benchmarks like CASAS (utilized by Alfaidi et al.) achieve high predictive accuracy (up to 94.2%) by leveraging dense sensor grids, yet their structural generalizability to diverse domestic environments remains limited. Conversely, wearable-centric large cohorts (such as Midroni et al. and Chan et al.) achieve strong predictive power for long-term dementia risk (C-index up to 0.729) but introduce compliance and missing-data challenges.
- **Multimodal Integration:** The literature clearly indicates that single-sensor approaches (such as isolated waist-worn IMUs) are reaching a performance ceiling (AUC $\approx 0.81 - 0.90$). Future breakthroughs depend heavily on the multimodal fusion of continuous ambient telemetry, wearable kinematic metrics, and clinical biomarkers, as modeled in the RADAR-AD framework and advanced deep-learning architectures (HEALNET).

Study	Technology	Design & Cohort	Key Metrics	Clinical Outcome
Park et al., 2023 [18]	Waist-worn IMU (single sensor)	Cross-sectional cohort (n = 595)	AUC: 0.811 (Validation)	Single-sensor gait analysis is a viable, repeatable marker for cognitive disorders.
Huang et al., 2026 [14]	Waist-worn IMU (SPPB tasks)	Cross-sectional case-control (n = 201)	AUC: 0.897, Accuracy: 82.8%	Scalable and interpretable pre-diagnostic screening tool using nonlinear features.
Heremans et al., 2025 [13]	Behind-the-ear wearable (EEG + accelerometry)	Cross-sectional proof-of-concept (n = 147)	AUC: 0.90 (AD detection)	One-channel EEG and accelerometry hold sufficient information for automated AD screening.
Dixit & Das, 2024 [8]	Consumer Smartwatches (IMU/PPG)	Simulation-based study (n = 6264 synthetic)	Accuracy: 94.4%	Proof-of-concept computational pipeline; requires validation on real human cohorts.
Yamasaki & Kumagai, 2021 [24]	In-home ambient sensors (PIR, contact switches)	Systematic Literature Review	ROC AUC: up to 0.97 (ML models)	Objective tracking of ADL anomalies serves as a reliable early digital biomarker for MCI.
Muurling et al., 2021 [17]	Multi-sensor RMTs (smartphones, wearables)	Multicentre observational study (n = 270 total)	Protocol & study design paper	Establishes a robust multi-sensor framework for real-world functional decline monitoring.
Alfaidi et al., 2026 [3]	Smart-home IoT (CASAS benchmark data)	Experimental data-driven framework	Accuracy: 94.2%	Multimodal IoT fusion provides reliable indicators for early cognitive-risk screening.
Akl et al., 2015 [2]	In-home passive infrared (PIR) sensors	Longitudinal observational study (n = 68)	AUC-ROC: 0.716 (12-week window)	Divergence from daily baselines detects the transition to MCI months prior to clinical diagnosis.
Chan et al., 2025 [6]	Wrist-worn accelerometer (Watch Walk)	Longitudinal prospective cohort (n = 47,371)	Cox proportional hazards models	Passively acquired gait and sleep biomarkers predict incident dementia comparably to traditional risk scores.
Midroni et al., 2024 [16]	Wrist accelerometry + Clinical features (Rush MAP)	Longitudinal predictive model (n = 1270)	Sensitivity: 89%, C-index: 0.729	Multimodal integration of wearables and clinical features improves five-year AD prediction sensitivity.

Table 1. Summary of Clinical Studies and Cohort Evaluations for Digital Health Technologies in Dementia and MCI Detection

7 CONCLUSION

This systematic literature review has examined the current landscape of digital health technologies for dementia care, bridging the gap between hardware-centric sensing infrastructures and advanced longitudinal predictive analytics.

The synthesized evidence demonstrates that continuous, non-invasive digital characterization, spanning passive smart-home IoT networks, wearable kinematic sensors, and multi-sensor remote monitoring platforms (such as RADAR-AD), holds profound potential for shifting dementia care from reactive, clinic-centric models to proactive, continuous, and naturalistic health management.

As revealed through our clinical study analysis, objective digital biomarkers derived from daily gait speed, step variability, sleep architecture, and room-level activity patterns can capture subtle behavioral anomalies months or even years prior to traditional clinical diagnoses. Frameworks like HEALNET and the Watch Walk methodology highlight how machine learning and deep learning pipelines can process these high-frequency telemetry streams to achieve predictive performance comparable to or exceeding established clinical risk scores, all while minimizing patient burden.

However, as underscored in our critical analysis, significant engineering and translational barriers remain. The deployment of these technologies faces persistent hurdles, notably the synthetic-versus-real-world data divide, domain-shift vulnerabilities in unconstrained environments, and severe adherence barriers driven by progressive cognitive decline. Furthermore, centralizing continuous behavioral telemetry introduces profound privacy and scalability challenges under stringent regulatory frameworks like GDPR.

Future research must prioritize validation on diverse, real-world clinical cohorts rather than benchmark or synthetic datasets. To ensure clinical translation, engineering efforts should pivot toward edge-computed federated learning, privacy-preserving differential privacy mechanisms, and context-aware sensor fusion. Overcoming these technical and compliance hurdles will be essential to establish trustworthy, scalable, and clinically actionable digital health solutions that can effectively alleviate the global burden of dementia care.

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